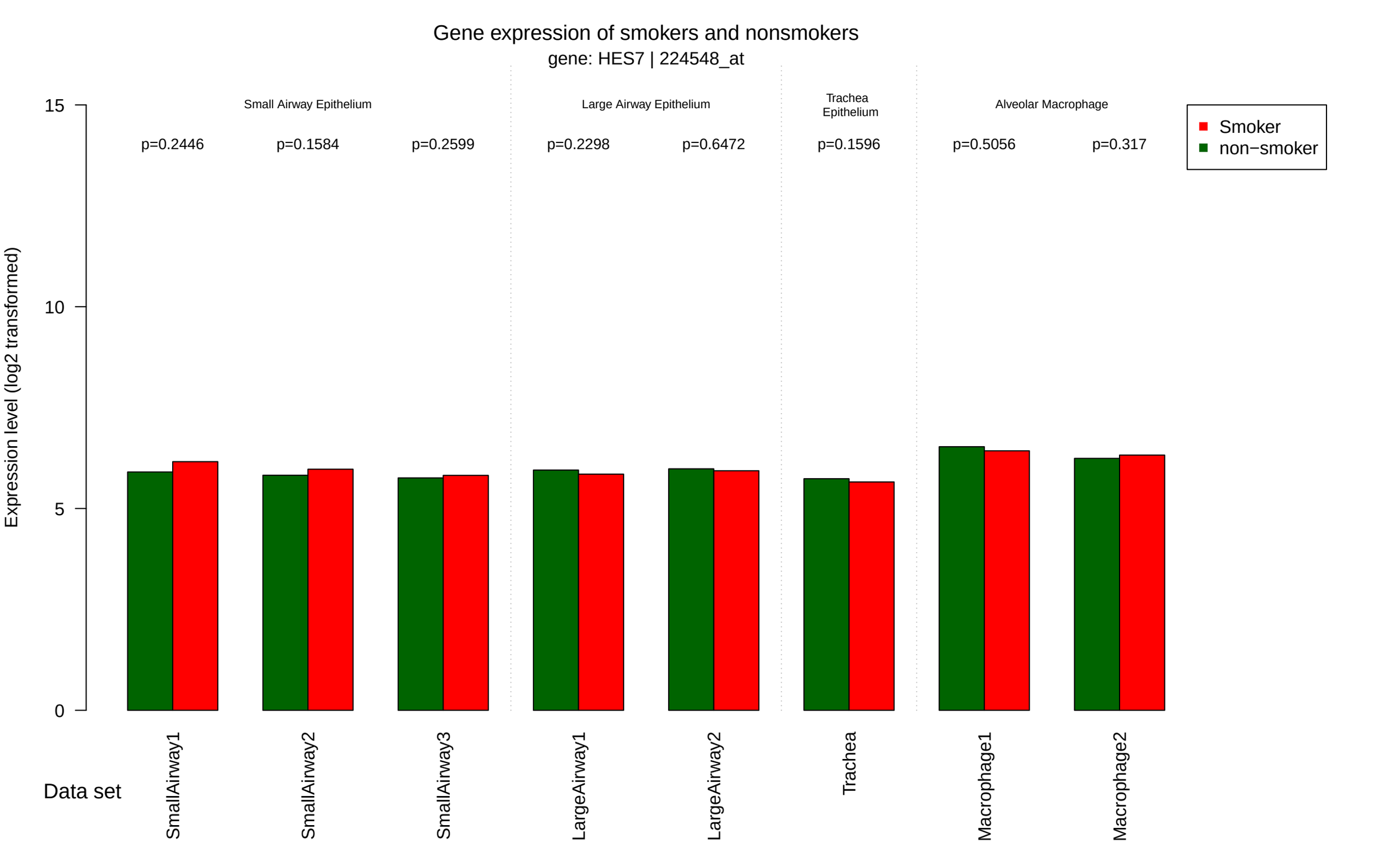
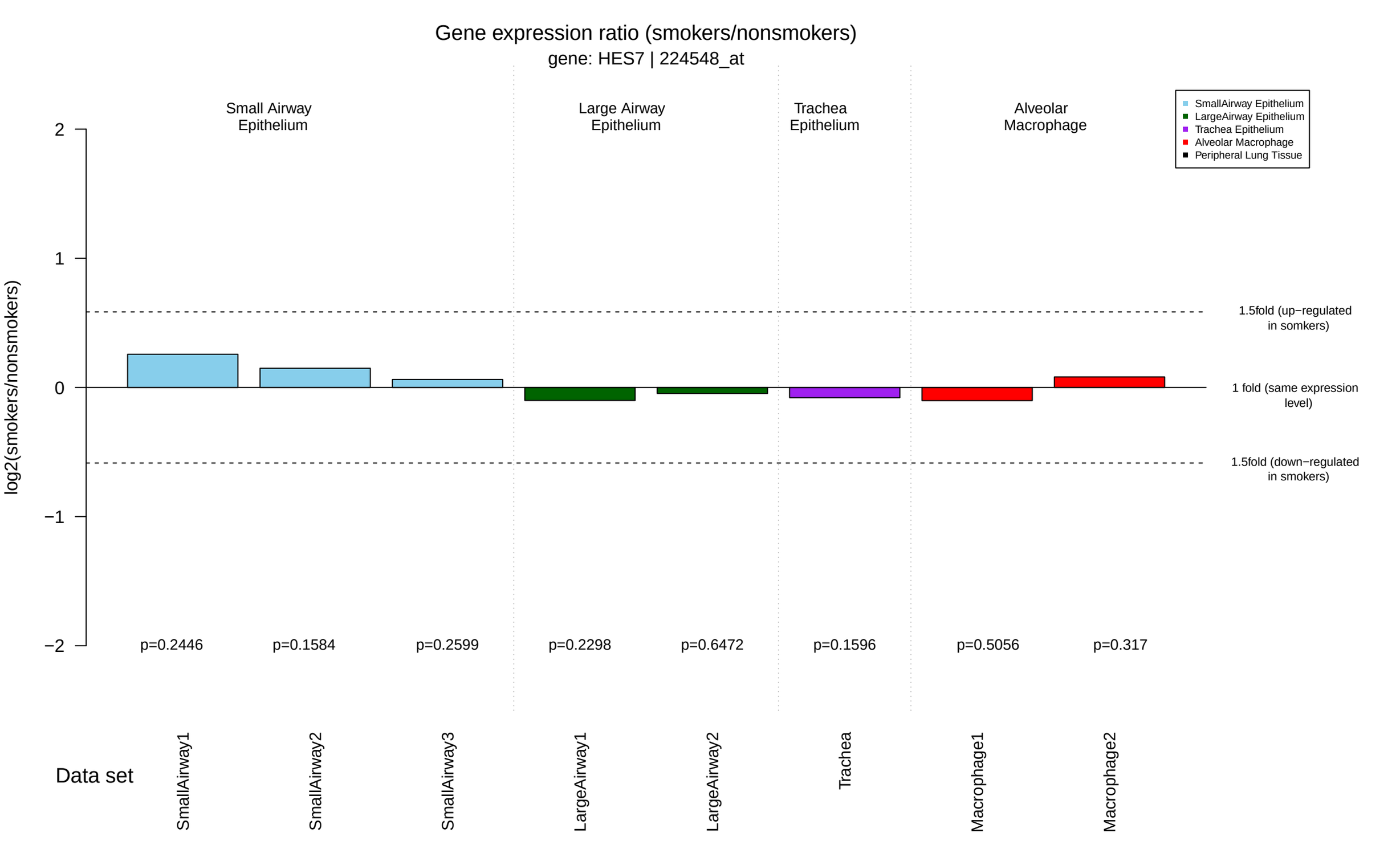
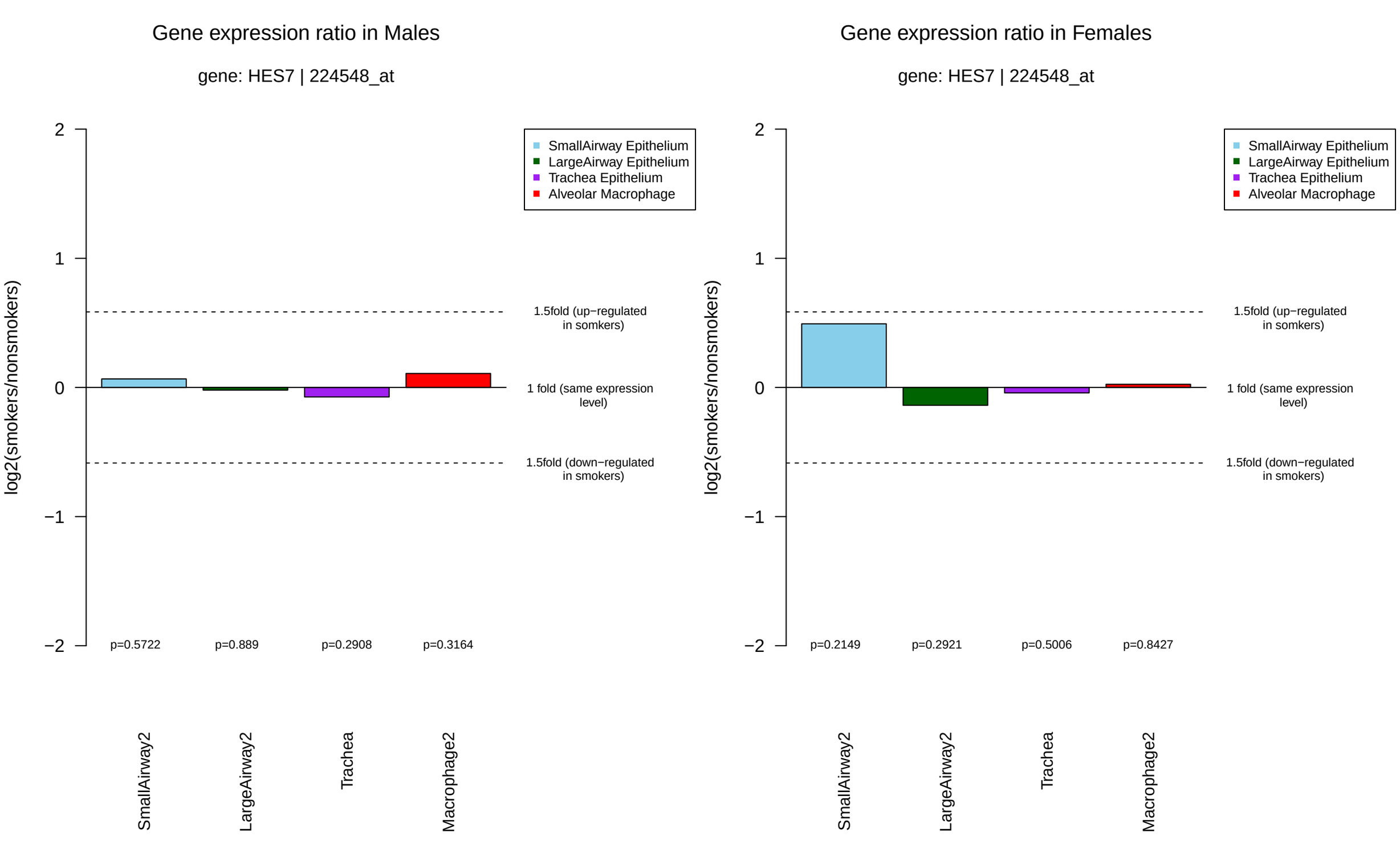
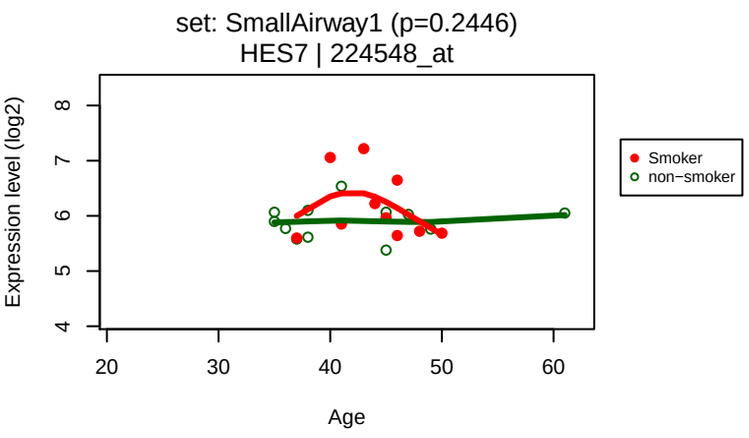
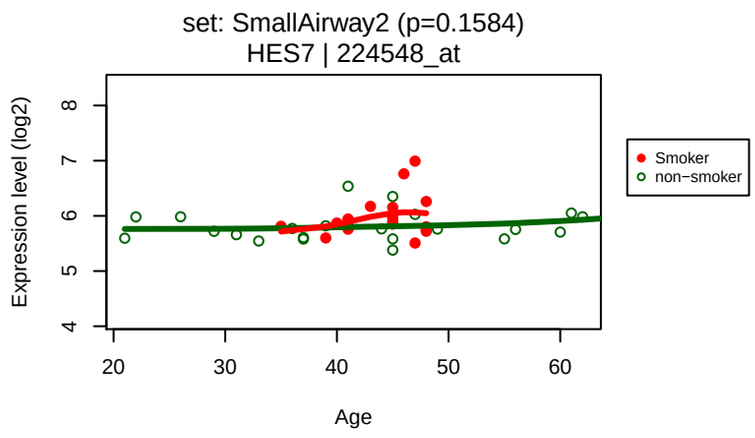
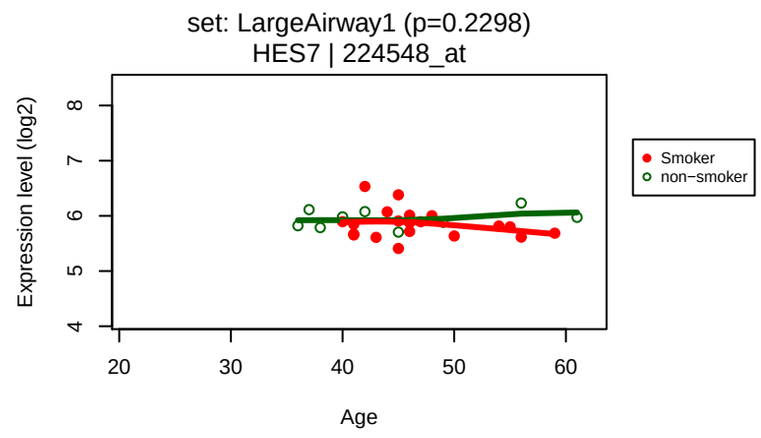

Gene expression of smokers and nonsmokers
gene: HES7 | 224548_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2446
p=0.1584
p=0.2599
p=0.2298
p=0.6472
p=0.1596
p=0.5056
p=0.317
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

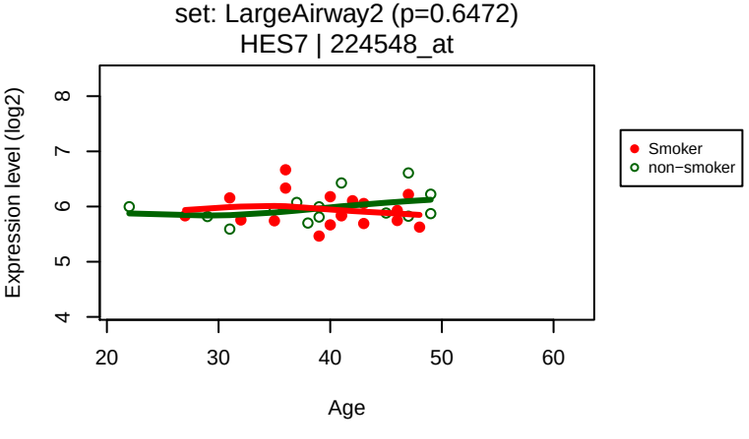

Gene expression ratio (smokers/nonsmokers)
gene: HES7 | 224548_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2446
p=0.1584
p=0.2599
p=0.2298
p=0.6472
p=0.1596
p=0.5056
p=0.317
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

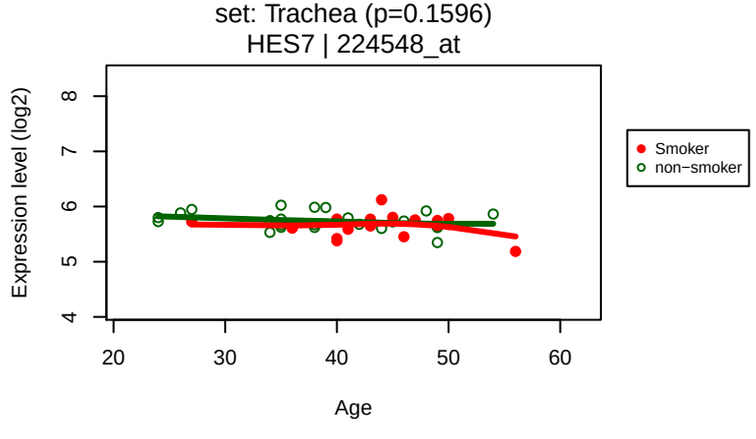

Gene expression ratio in Males
gene: HES7 | 224548_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5722
p=0.889
p=0.2908
p=0.3164
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HES7 | 224548_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2149
p=0.2921
p=0.5006
p=0.8427
SmallAirway2
LargeAirway2
Trachea
Macrophage2

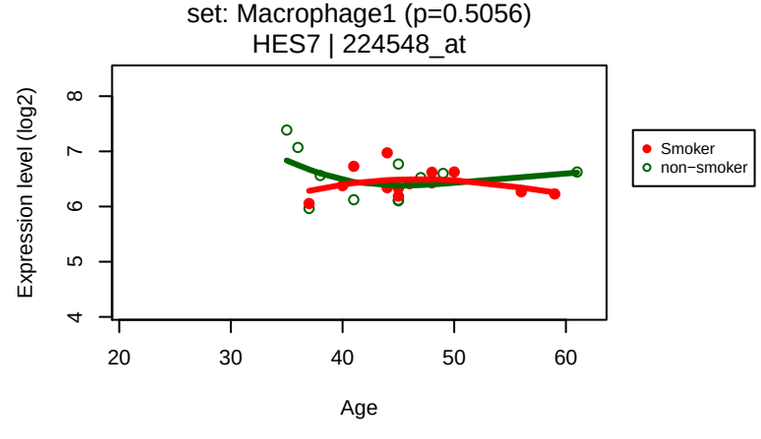

set: SmallAirway1 (p=0.2446)
HES7 | 224548_at
Expression level (log2)
Age
Smoker
non−smoker

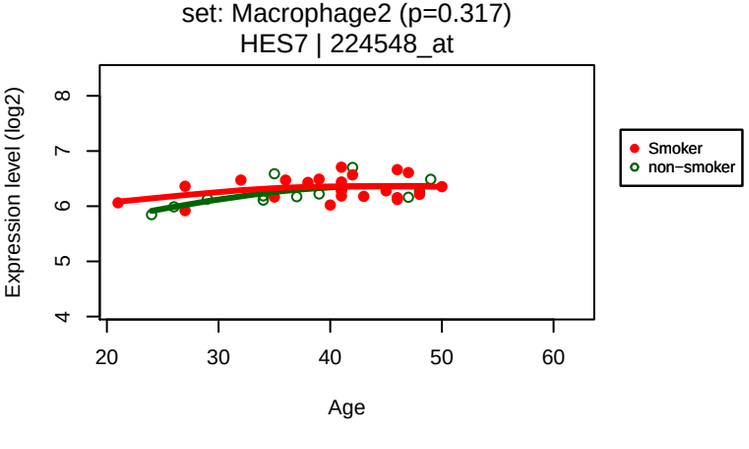

set: SmallAirway2 (p=0.1584)
HES7 | 224548_at
Expression level (log2)
Age
Smoker
non−smoker

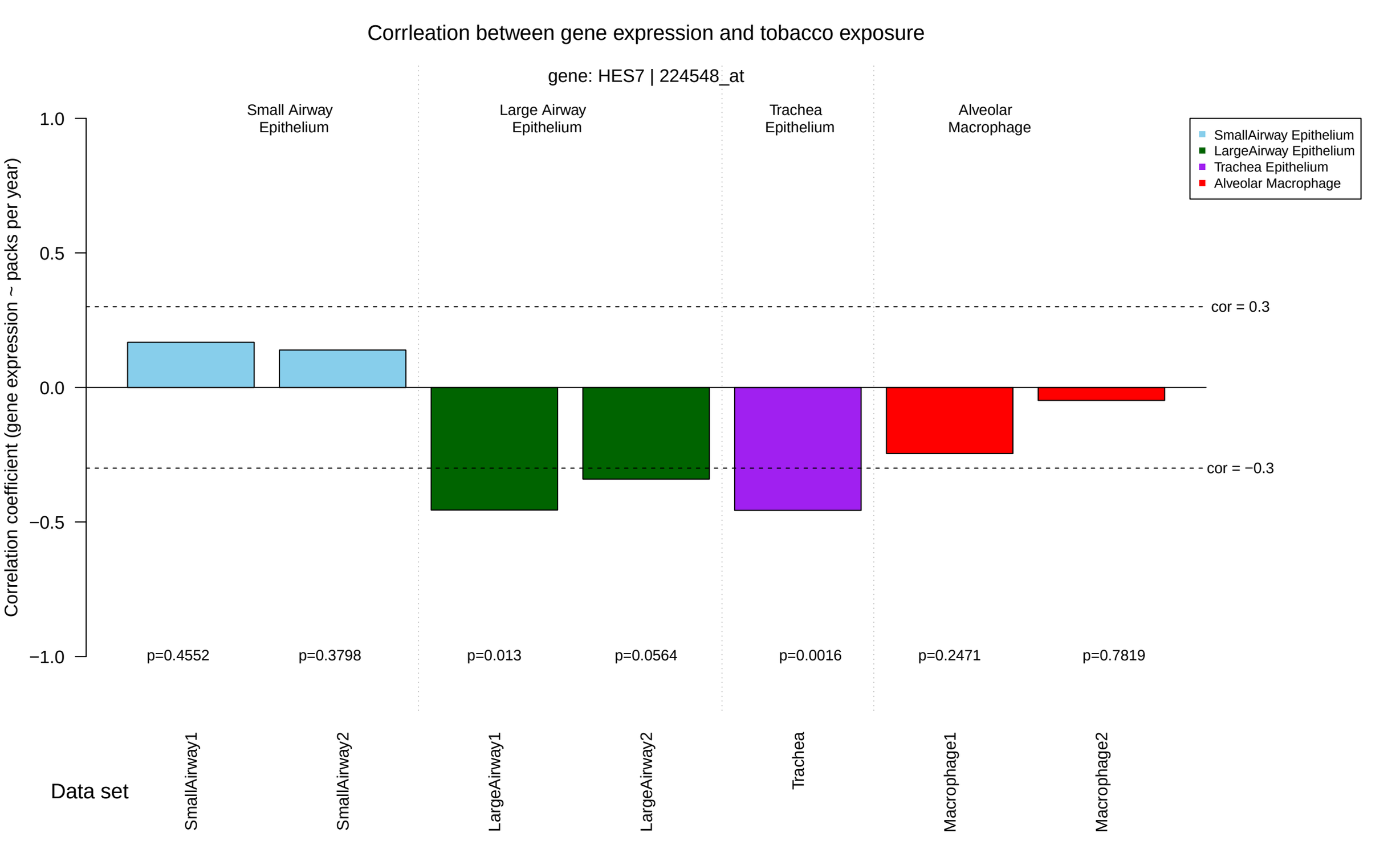

set: LargeAirway1 (p=0.2298)
HES7 | 224548_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6472)
HES7 | 224548_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1596)
HES7 | 224548_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5056)
HES7 | 224548_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.317)
HES7 | 224548_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HES7 | 224548_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4552
p=0.3798
p=0.013
p=0.0564
p=0.0016
p=0.2471
p=0.7819
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2